

Lesson 28 Notes:

1. General Pattern with `bokeh.plotting`

Typical Bokeh workflow:

1. Import and create a **figure**.
2. Add one or more **glyphs** (line, circle, vbar,...).
3. Call `show()`.

```
from bokeh.io import output_notebook
from bokeh.plotting import figure, show
```

```
output_notebook() # for Jupyter
```

```
p = figure(height=300, width=300)
p.line([1, 2, 3, 4, 5], [6, 7, 2, 1, 5])
show(p)
```

2. Line Plots

2.1 Simple line

```
from bokeh.plotting import figure, show
```

```
p = figure(height=300, width=300, title="Basic Line Plot")
x = [1, 2, 3, 4, 5]
y = [6, 7, 2, 1, 5]
```

```
p.line(
    x, y,
    legend_label="Series 1",
```

```
    line_width=2,  
    line_color="red",  
)  
  
show(p)
```

2.2 Line chart from a DataFrame

```
import pandas as pd  
from bokeh.plotting import figure, show  
  
# passengers per month in 2021  
months = list(range(1, 13))  
passengers_in_millions = [35.1, 38.4, 45.2, 51.0, 58.9, 63.1, 65.4, 64.9, 59.3, 55.1, 52.8, 54.6]  
  
monthly_values_df = pd.DataFrame(  
    data={"passengers": passengers_in_millions},  
    index=months  
)  
monthly_values_df.index.name = "month"  
  
p = figure(  
    height=300,  
    width=600,  
    title="Domestic Airline Passengers (2021)",  
)  
  
x = monthly_values_df.index  
y = monthly_values_df["passengers"]  
  
p.line(x, y, legend_label="Passengers", line_width=3, line_color="blue")  
show(p)
```

2.3 Line + markers

```

from bokeh.plotting import figure, show

p = figure(height=300, width=300)
x = [1, 2, 3, 4, 5]
y = [6, 7, 2, 1, 5]

p.line(x, y)
p.scatter(x, y, size=15, color="red", alpha=0.5, marker="circle_dot")

show(p)

```

3. Scatter Plots and Marker Types

```

from bokeh.plotting import figure, show

p = figure(width=400, height=200, title="Scatter Example")
p.scatter([1, 2, 3, 4, 5], [6, 7, 2, 4, 5], size=15) # default circle

show(p)

```

You can change markers using `marker` (e.g. `"circle"`, `"square"`, `"triangle"`, `"asterisk"`), or use specific glyph methods like `p.circle`, `p.square`, etc.

4. Categorical Bar Charts with `vbar`

4.1 Simple bar chart

```

from bokeh.plotting import figure, show

fruits = ["Apples", "Pears", "Nectarines", "Plums", "Grapes", "Strawberries"]
counts = [5, 3, 4, 2, 4, 6]

p = figure(

```

```

x_range=fruits,      # categorical x-axis
height=350,
title="Fruit Counts",
)

p.vbar(x=fruits, top=counts, width=0.9, color="firebrick")

show(p)

```

4.2 Bar chart from a DataFrame

```

import pandas as pd
from bokeh.plotting import figure, show

top_airlines = {
    "unique_carrier_name": [
        "Southwest Airlines", "American Airlines", "Delta Air Lines",
        "United Airlines", "SkyWest Airlines", "Alaska Airlines",
        "Spirit Airlines", "JetBlue Airways", "Frontier Airlines", "Republic Airline"
    ],
    "passengers": [11.5, 10.8, 9.9, 9.1, 7.6, 5.4, 5.1, 4.8, 3.9, 3.5]
}
airlines_by_passengers_df = pd.DataFrame(top_airlines)

p = figure(
    x_range=airlines_by_passengers_df["unique_carrier_name"][:10],
    title="Top 10 carriers by passengers (domestic routes)",
    height=300,
    width=600,
)

p.vbar(
    x=airlines_by_passengers_df["unique_carrier_name"],
    top=airlines_by_passengers_df["passengers"],
    width=0.6,
)

```

```
    legend_label="Passengers",
)

p.xaxis.major_label_orientation = 0.8 # rotate x labels
show(p)
```

5. Hover Tool (Interactive Tooltip)

```
from bokeh.plotting import figure, show
from bokeh.models import ColumnDataSource, HoverTool

source = ColumnDataSource(
    data=dict(
        x=[1, 2, 3, 4, 5],
        y=[2, 5, 8, 2, 7],
        desc=["A", "b", "C", "d", "E"],
    )
)

p = figure(height=300, title="Hover Tool Example")
p.circle(x="x", y="y", size=20, source=source)

hover = HoverTool(
    tooltips=[
        ("index", "$index"),
        ("(x, y)", "(@x, @y)"),
        ("label", "@desc"),
    ]
)
p.add_tools(hover)

show(p)
```

Hover tools read values from the **ColumnDataSource** via `@column_name` .

6. **ColumnDataSource** : Bokeh's Core Data Structure

6.1 Concept

- Maps **column names** → **sequences of values** (lists, NumPy arrays, Series).
- Most glyphs can use `source=...` and refer to columns by name.

6.2 From a dictionary

```
from bokeh.models import ColumnDataSource

source = ColumnDataSource(
    data={
        "x": [1, 2, 3, 4, 5],
        "y": [3, 7, 8, 5, 1],
    }
)

source.data["x"] # access a column
```

Lists (or arrays) in each column **must have the same length**.

6.3 From Series + NumPy arrays

```
import numpy as np
import pandas as pd
from bokeh.models import ColumnDataSource

passengers = np.random.randint(100, 500, size=12)
monthly_passengers_series = pd.Series(passengers, name="passengers")
range_array = np.arange(len(monthly_passengers_series))

source = ColumnDataSource(
    data={
        "x": monthly_passengers_series,
        "y": range_array,
```

```
}  
)
```

6.4 Using `ColumnDataSource` in glyphs

```
from bokeh.plotting import figure, show  
from bokeh.models import ColumnDataSource  
  
data_dict = {"x_values": [1, 2, 3, 4, 5], "y_values": [6, 7, 2, 3, 6]}  
source = ColumnDataSource(data=data_dict)  
  
p = figure(height=300)  
p.scatter(  
    x="x_values",  
    y="y_values",  
    source=source,  
)  
  
show(p)
```

7. ColumnDataSource from a DataFrame

```
import pandas as pd  
import numpy as np  
from bokeh.models import ColumnDataSource  
  
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',  
          'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']  
  
data = {  
    'month_name': months,  
    'passengers': np.random.randint(1000, 5000, size=12),  
    'freight': np.random.randint(500, 2000, size=12),  
    'mail': np.random.randint(100, 500, size=12),
```

```
}  
monthly_values_df = pd.DataFrame(data)  
  
source = ColumnDataSource(monthly_values_df)
```

7.1 Stacked bar chart with `vbar_stack`

```
from bokeh.plotting import figure, show  
from bokeh.palettes import HighContrast3  
  
p = figure(  
    x_range=source.data["month_name"],  
    width=750,  
    title="Monthly Values (Stacked)",  
)  
  
columns = ["freight", "passengers", "mail"]  
  
p.vbar_stack(  
    columns,  
    x="month_name",  
    width=0.9,  
    source=source,  
    color=HighContrast3,  
    legend_label=columns,  
)  
  
p.legend.location = "top_left"  
show(p)
```

8. Transformations: `cumsum` and `linear_cmap`

8.1 Cumulative bar chart with `cumsum`


```

from bokeh.plotting import figure, show
from bokeh.models import ColumnDataSource
from bokeh.transform import cumsum

source = ColumnDataSource(monthly_values_df)

p = figure(
    height=300,
    x_range=source.data["month_name"],
    title="Cumulative Passengers",
)

p.vbar(
    x="month_name",
    top=cumsum("passengers", include_zero=True),
    width=0.9,
    source=source,
)

show(p)

```

8.2 Color mapping with `linear_cmap`

```

import numpy as np
from bokeh.plotting import figure, show
from bokeh.transform import linear_cmap

N = 2000
source = dict(
    x=np.random.random(size=N) * 100,
    y=np.random.random(size=N) * 100,
    r=np.random.random(size=N) * 1.5,
)

```

```

p = figure(height=300, title="Color Mapped Scatter")

p.circle(
    x="x",
    y="y",
    radius="r",
    source=source,
    fill_alpha=0.6,
    color=linear_cmap("x", "Viridis256", 0, 100), # color by x value
)

show(p)

```

9. Example: Categorical Bar Chart with Hover

```

from bokeh.plotting import figure, show
from bokeh.models import HoverTool

fruits = ['Apples', 'Pears', 'Nectarines', 'Plums', 'Grapes']
counts = [5, 3, 4, 2, 6]

p = figure(
    x_range=fruits,
    height=350,
    title="Fruit Counts with Hover",
)

p.vbar(x=fruits, top=counts, width=0.9)

hover = HoverTool(
    tooltips=[
        ("Fruit", "@x"),
        ("Count", "@top"),
    ]
)

```

```
)  
p.add_tools(hover)  
  
p.xgrid.grid_line_color = None  
p.y_range.start = 0  
  
show(p)
```
